



GREEN PUF

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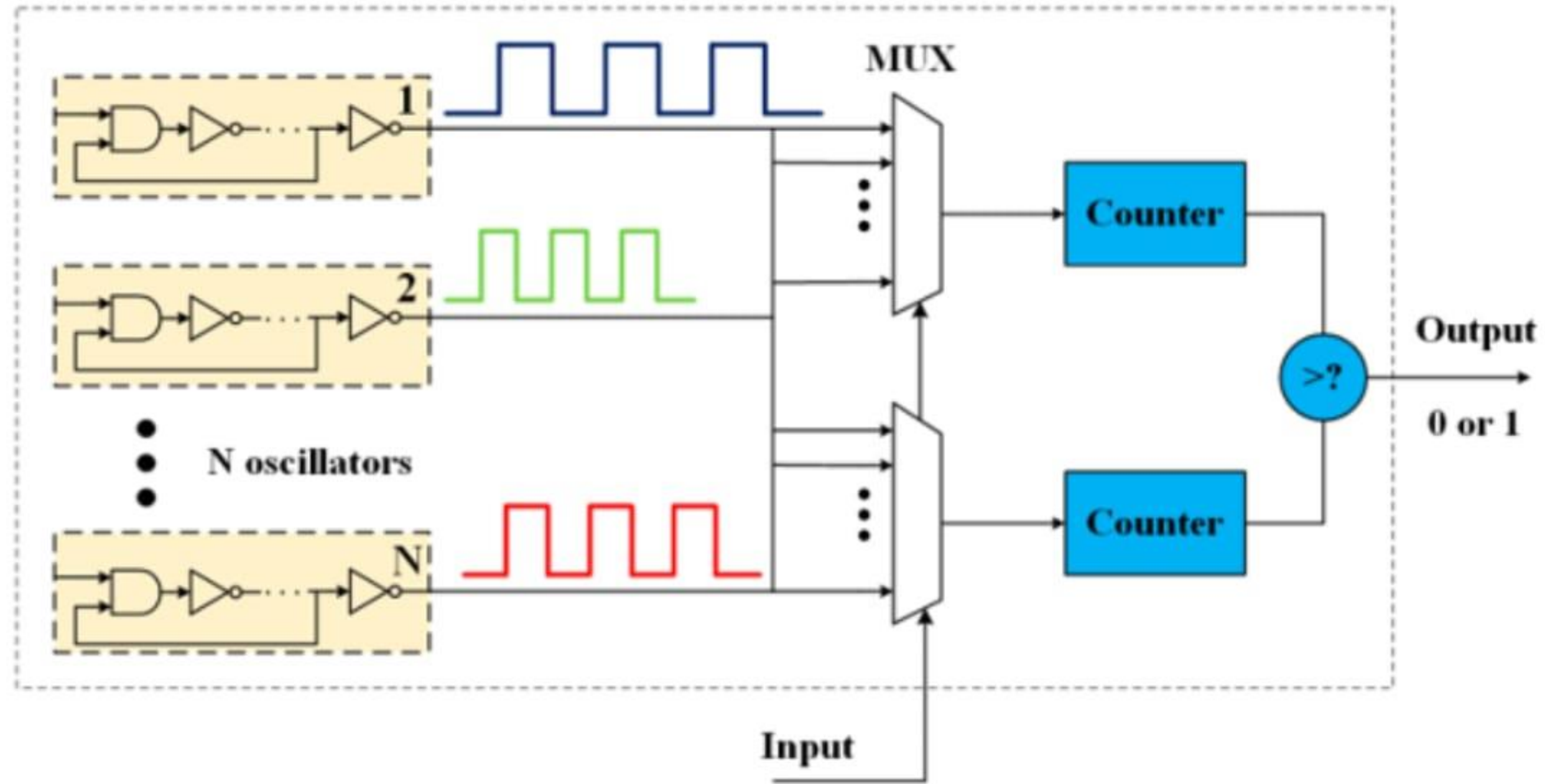
Nitte Meenakshi Institute Of Technology

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Overview

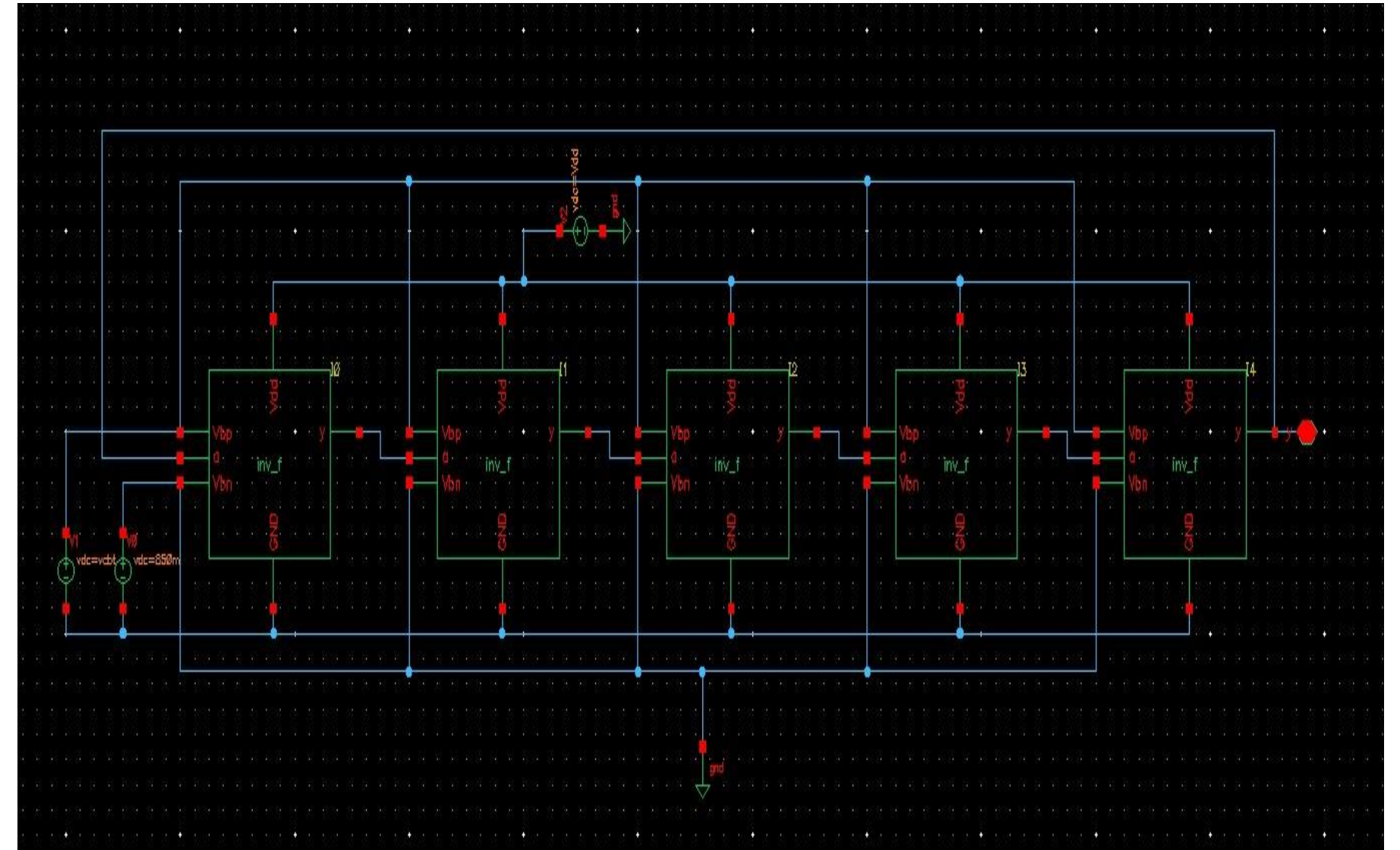
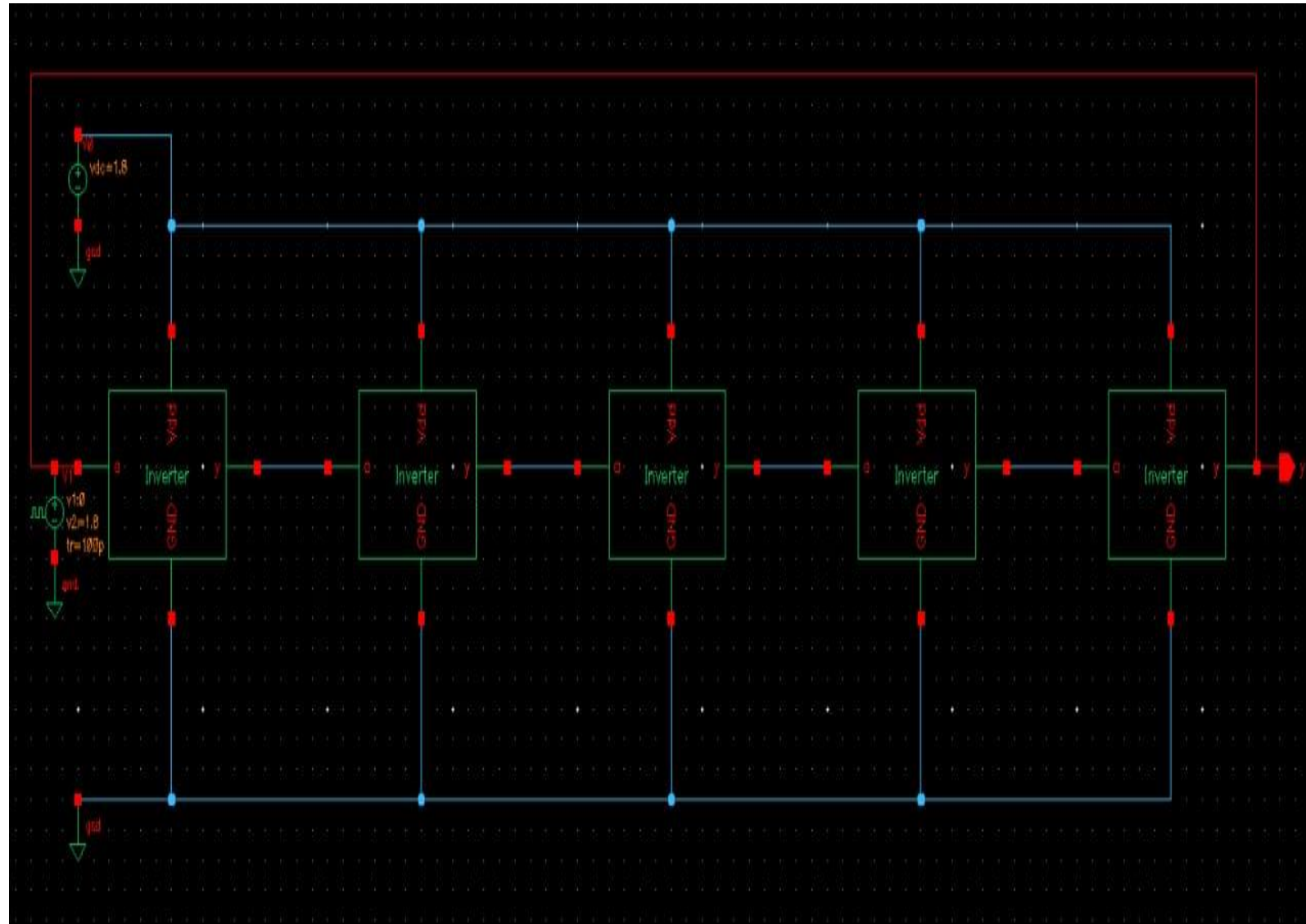
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- 03 Challenges and Limitations
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Background



- **IoT Boom:** Billions of devices need secure, low-energy authentication
- **Problem with Traditional PUFs:** High power, unstable with temperature, rely on bulky ECC
- **Our Goal:** Design a “Green PUF” that is low-power, reliable, and secure without ECC overhead

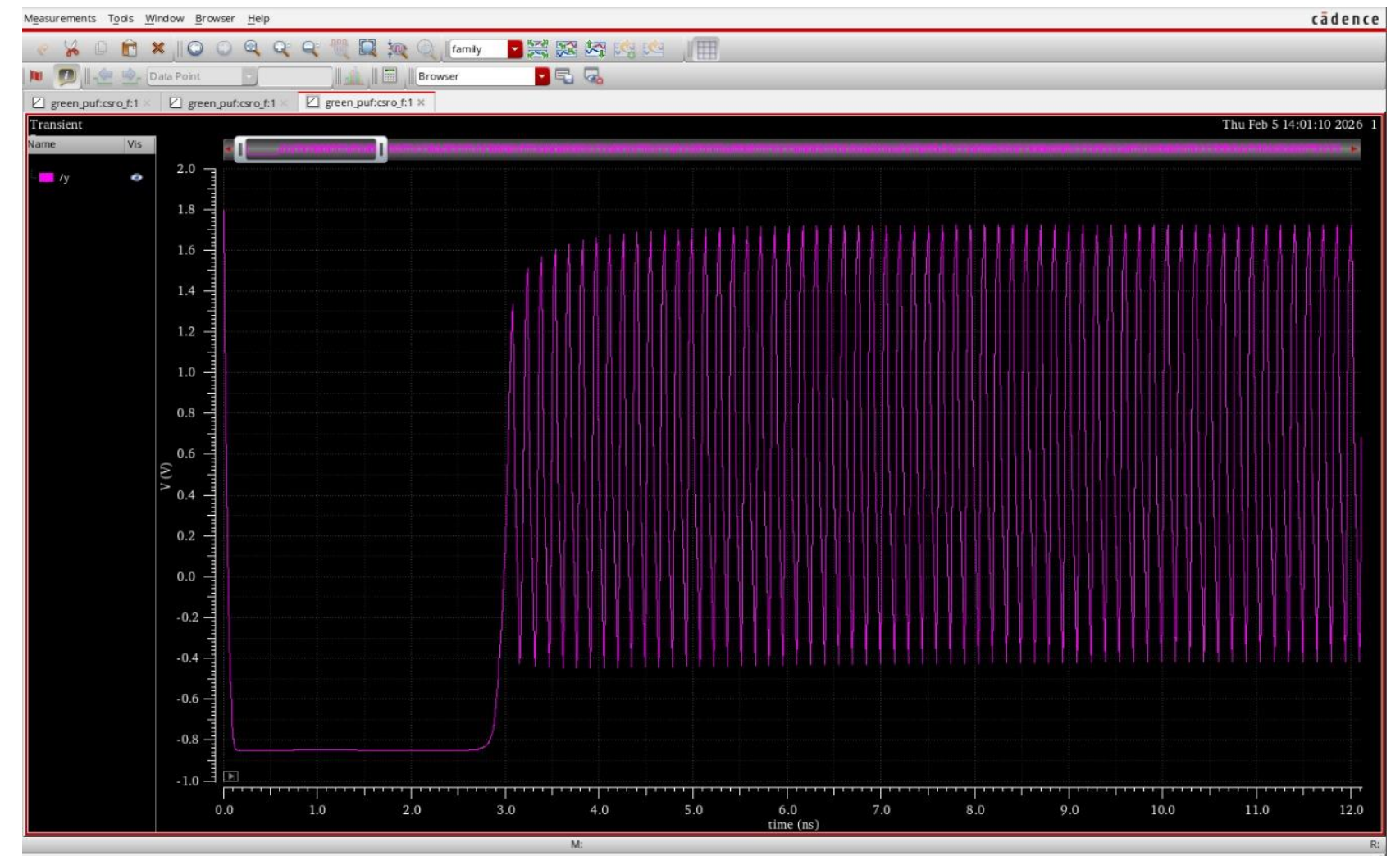
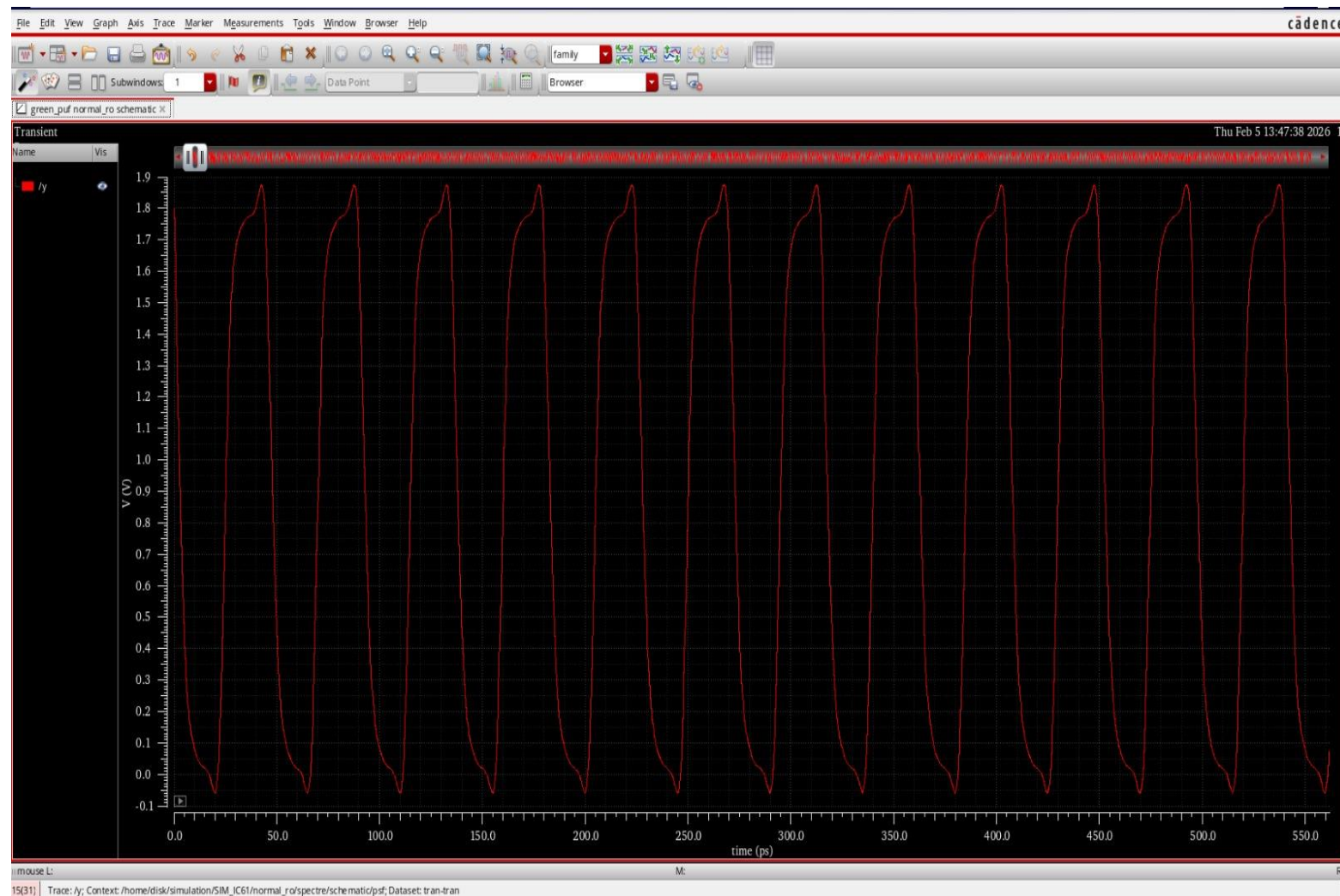
Concept I: Circuit Diagram



Circuit Diagram and Simulation Results

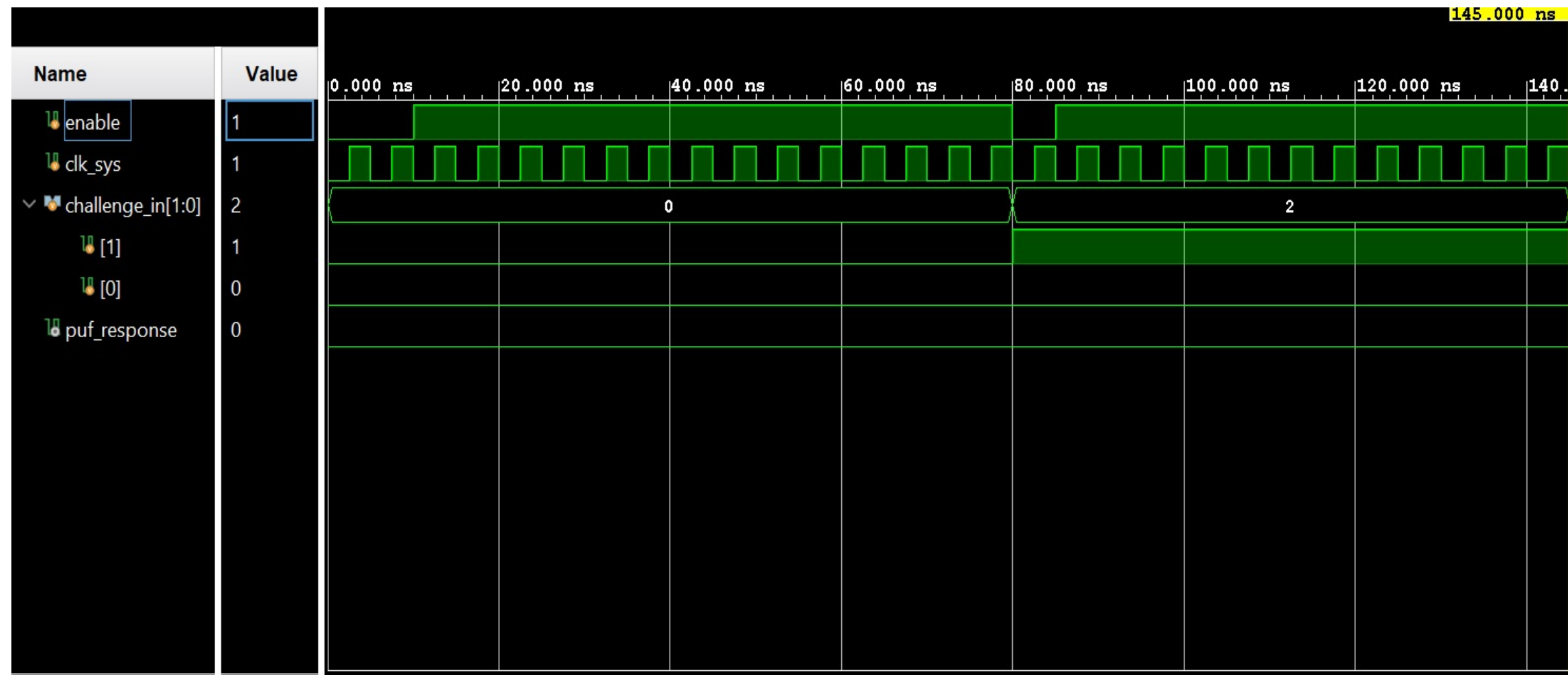
- **Normal RO:** Inverters connected directly to VDD/GND → always drawing maximum current.
- **CSRO:** Adds a transistor (header/footer) that limits current flow → like a "water tap" controlling flow.

Simulation Results



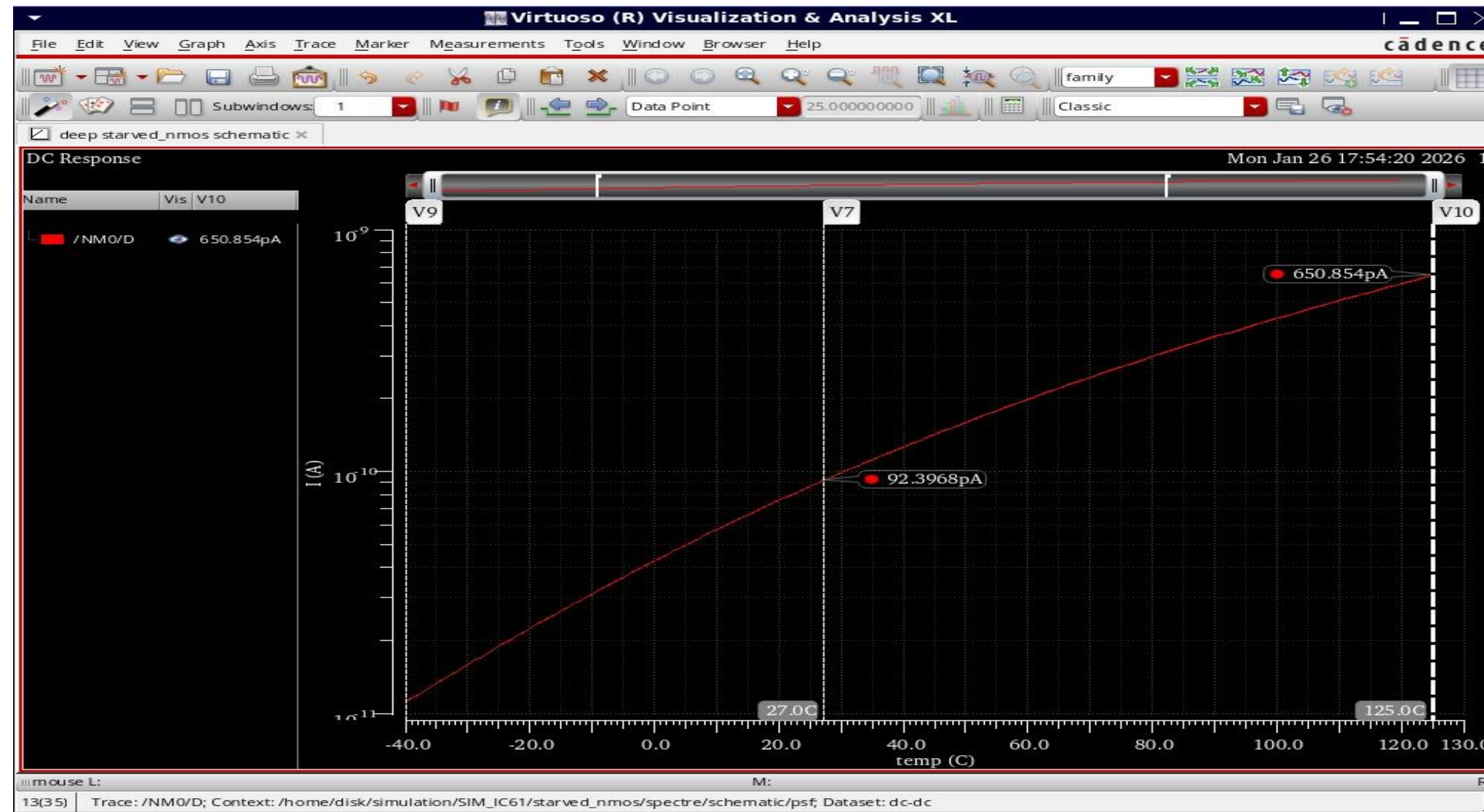
- The first image indicates the output of the normal RO (Ring oscillator)
- The second image indicates the output of the CSRO (Current starved ring oscillator)

RTL Simulation



- Simulation output of a complete PUF response, which consists of four, 5 stage CSRO's with varying frequencies of 2% difference

Challenges/Limitations



Temperature Sweep

Temperature Sensitivity:

- For a lower bias voltage, after running the temperature sweep, we found that the Drift ratio was much higher than expected.
- This made the current very unstable at different temperatures

Challenges/Limitations-II (Optional)



- To find the w\l ratio, we did parametric analysis, and our biggest challenge was to find the sweet spot to balance out the CSRO design

Summary

- **Green-PUF** = Low power + High reliability + Strong security
- **Impact:** Enables sustainable IoT security (wearables, medical devices)
- **Next Steps:**
 - a) Silicon tape-out
 - b) Integration into IoT SoC
 - c) Expand LNN for voltage variation tolerance